

Research article

Assessment of spatial variability of multiple ecosystem services in grasslands of different intensities



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ABSTRACT

Grasslands provide multiple Ecosystem Services (ES) such as forage provision, carbon sequestration or habitat provision. Knowledge about the trade-offs between these ES is of great importance for grassland management. Yet, the outcome of different management strategies on ES provision is highly uncertain due to spatial variability. We aim to characterize the provision (level and spatial variability) of grassland ES under various management strategies. To do so, we combine empirical data for multiple ES with spatially explicit census data on land use intensities. We analyzed the variations of five ES (forage provision, climate regulation, pollination, biodiversity conservation and outdoor recreation) using data from biodiversity fieldwork, experimental plots for carbon as well as social network data from Flickr. These data were used to calculate the distribution of modelled individual and multiple ES values from different grassland management types in a Swiss case study region using spatial explicit information for 17,383 grassland parcels. Our results show that (1) management regime and intensity levels play an important role in ES provision but their impact depends on the ES. In general, extensive management, especially in pastures, favors all ES but forage provision, whereas intensive management favors only forage provision and outdoor recreation; (2) ES potential provision varies between parcels under the same management due to the influence of environmental drivers, related to topography and landscape structure; (3) there is a trade-offs between forage provision and other ES at the cantonal level but a synergy between forage provision and biodiversity conservation within the grassland categories, due to the negative impact of elevation on both ES. Information about multiple ES provision is key to support effective agri-environmental measures and information about the spatial variability can prevent uncertain outputs of decision-making processes.

1. Introduction

Grasslands cover approximately 40% of the Earth's land surface (Blair et al., 2014). These grasslands provide multiple Ecosystem Services (ES), ranging from provisioning services, regulating services and cultural services (Allan et al., 2015; Baldocchi et al., 2017; Harrison et al., 2010). The provision of these ES, and their interrelations, are highly affected by grassland management regimes and intensity (Jeanneret et al., 2007). Intensively used grasslands provide higher forage quantity and quality (Beckmann et al., 2019; Qi et al., 2018) while low-intensive grasslands are associated with higher biodiversity

and related ES (Marini et al., 2008). In addition to management factors, environmental factors such as soil quality or elevation can affect ES provision (Feng et al., 2017; Wang et al., 2007).

Interrelationships among ES may be affected by the spatial allocation of grassland management regimes and intensities (Wu et al., 2017). Understanding the associations of ES coming from different grassland regimes and intensities in space would allow to identify areas in which production and biodiversity can be advanced jointly (Simons and Weisser, 2017). Moreover, leverage points can be identified where small management investments can yield substantial benefits (Bennett et al., 2009). Thus, better knowledge about relationships among ES in

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grasslands represents an important source for the design of natural resource management approaches in coupled human and natural systems (Kramer et al., 2017; Manning et al., 2018) and may contribute to counter ongoing land use intensification that reduces biodiversity (Allan et al., 2015; Birkhofer et al., 2018).

Research on ES associations (Mouchet et al., 2014) or bundles of ES (Spake et al., 2017) often treats grassland as one single type (e.g. Raudsepp-Hearne et al., 2010). This, however, is not always enough to be used for effective environmental management (Van der Biest et al., 2015). Moreover, nearly all assessments to date do not consider ES provision on the plot which is ultimately the spatial level where farmers' decide on the management regime and intensity (Verhagen et al., 2018). This is of specific relevance for Europe, where agri-environmental schemes incentivize low intensive grassland use.

In this paper, we aim to contribute filling these gaps in the literature. We characterize the spatial provision of grassland ES in a multifunctional agricultural landscape under various grassland categories or management strategies (17,383 parcels of meadows or pastures, both at different intensities). We quantify the extent and variability if the provision of ES across grassland categories and quantify how variable the ES provision is within each category. More specifically, the research questions are: i) What is the spatial variability of multiple grassland ES in a multifunctional agricultural landscape? and ii) What are the trade-offs and synergies resulting from different grassland categories (based on management regimes and intensities)? A better understanding of ES associations resulting from different management regimes and intensities on parcel level can result in policies that are more efficient. The remainder of this paper is organized as follows. In the next Section, we describe the background of our study including the definition of grassland management regimes and intensities and of the spatial variability in ES provision and the policy context. In Section 3, we present the case study region, the outline of our methodology, followed by the data we used. In Section 4, we present the results of our analysis. Finally, we discuss the implications for agricultural policy.

2. Background: policy context and definitions of main concepts

2.1. Agri-environmental measures to support grassland ES

Environmental goals have become an important pillar in European agricultural policies (Matthews, 2013). The maintenance of existing permanent grassland and the increase of grassland under low-intensity through agri-environmental measures are of specific relevance in this context. Permanent and low-intensive grasslands are expected to support many ES such as biodiversity conservation, carbon sequestration and landscape maintenance (Uthes and Matzdorf, 2013). However, the effectiveness of these measures is low (Pe'er et al., 2014) and grassland management plays an important role in further improving the environmental performance of the European Common Agricultural Policy (Navarro and López-Bao, 2018). In Switzerland many policy instruments incentivize low-intensive grassland use (Huber et al., 2017). Agri-environmental schemes for the extensive management of pastures and meadows help to conserve biodiversity in multifunctional agricultural landscapes (Herzog et al., 2005; Kampmann et al., 2012). Despite these efforts, the European and Swiss agricultural policy does not achieve its environmental targets. All these policy measures focus on the intensity of grassland management and its spatial allocation in the landscape. Thus, empirical based information about the trade-offs between grassland ES emerging from the plot based spatial variability and the choice of the grassland regime and intensity helps informing environmental management and the design of agri-environmental policy schemes in the European context (Engel, 2016).

2.2. Associations of ES in grassland categories

To clarify the underlying concepts used in this article, we define in

the following paragraphs grassland categories, bundles of grassland ES, trade-offs and synergies, hotspots and coldspots as well as spatial variability of ES provision. Definitions on ecosystem service concepts are based on recent literature reviews by Mouchet et al. (2014) (Spake et al., 2017), and (Frei et al., 2018).

Grassland categories: Grassland can be managed with different regimes i.e., pastures or meadows, and with different intensity levels i.e., more or less fertilizer, number of cuts and or livestock density. These different grassland categories provide different levels of ES and biodiversity (Beckmann et al., 2019). More intensively used grasslands i.e., fertilized grassland with multiple cuts or a high stocking density provide higher forage quantity and quality (Beckmann et al., 2019; Qi et al., 2018). In contrast, low-intensive grasslands are expected to have higher biodiversity and related ES (Bengtsson et al., 2019; Habel et al., 2013; Marini et al., 2008). In this study, we considered five grassland categories: intensive, less intensive and extensive meadows as well as intensive and extensive pastures (see subsection 3.3).

Bundles of grassland ES: Bundles refer to “sets of ecosystem services that repeatedly appear together across space or time” (Raudsepp-Hearne et al., 2010). We define bundle as set of grassland ES that is provided repeatedly in space by our grassland categories. This is of fundamental importance since most of the European agricultural policies focus on the management of plots i.e., our grassland categories and not the management of landscapes or ecosystems. In line with Vannier et al. (2019), we argue that understanding ES associations beyond broad land-use or land-cover classes is key in gaining information for environmental management.

Trade-offs and synergies in grassland ES: Trade-offs “occur when the provision of one ES is reduced as a consequence of increased use of another ES” (Rodríguez et al., 2006). Synergies arise when the use of an ES simultaneously increases another ES (Spake et al., 2017). We here show the positive or negative association that exist across and within grassland categories and how they relate to underlying environmental factors. We focus on supply-supply relationship in different grassland categories and do not consider any further socio-economic aspects (see Mouchet et al., 2014 for other relationships).

Hotspots and coldspots: Hotspots and coldspots are areas providing, respectively, high and low amounts of one or several ES (Schulp et al., 2014).

Spatial variability: Spatial levels play an important role in the assessment of the relationships between ES (Qiu et al., 2018; Raudsepp-Hearne and Peterson, 2016; Simons and Weisser, 2017; Vallet et al., 2018). Underlying environmental factors can cause two types of variability. First, there is a spatial variability on the parcel level. Because of structural differences across or within parcels (e.g. soil types, elevation and average climatic conditions), the provision of ES such as grassland yields (Huguenin-Elie et al., 2017), pollination (Dauber et al., 2003) or plant species richness (Bruun, 2001) can differ substantially – even under similar management. Secondly, there is a spatial variability from the landscape composition and structure. For example, ES provision depends also on the characteristics of neighboring plots and surrounding landscape (Duflot et al., 2017; Hendrickx et al., 2007; Le Feon et al., 2010; Reitalu et al., 2009; Tschardt et al., 2012) or recreational values (Colson et al., 2010; Edwards et al., 2012; Gul et al., 2006). We include both of these sources of variability in our analysis and define spatial variability as the positive or negative deviation from the mean ES provision per grassland category (Table 1). Thus, grassland ES provision does not only depend on the management strategy but also on the underlying, spatially explicit environmental factors.

3. Material and methods

3.1. Case study area

We used the Swiss Canton of Solothurn as a case study region (Fig. 1). Solothurn is located in the northwest of Switzerland and covers

Table 1

Sources of spatial variability for ES in parcels under the same grassland category. “Not available” means that consideration of the uncertainty was not possible with the dataset we used.

ES (indicator)	Spatial variability	
	Parcel	Landscape structure
Forage provision (yield)	Considered (elevation and soil)	Not available
C sequestration (composite variable)	Considered (elevation)	Not available
Pollination (bee richness)	Considered (diverse environmental factors)	Considered
Biodiversity conservation (plant richness)	Considered (diverse environmental factors)	Considered
Outdoor recreation (photos taken in grassland area)	Considered (elevation)	Not available

an area of 791 km². It presents a wide range of elevations from the plain created by the Aare River (277 m.a.s.l) to the foothills of the Jura massif (1,445 m.a.s.l). For centuries, agriculture has been the dominant land use in the canton. Agricultural land use is characterized by small-scale and diversified farming systems. Average farm size in the Canton of Solothurn is 23 ha and average parcel size is 0.9 ha, resulting in a heterogeneous pattern of croplands and grasslands. The predominant agricultural land use is permanent grasslands that covered around 165 km², i.e., 67% of the agricultural area in 2015, while rotational grasslands and cropland cover 14% and 32% of the cantonal area, respectively (FSO, 2015). 1,170 farms cultivated 17,383 parcels of

permanent grasslands. Two third of these grasslands were meadows and one third was used as pastures. The dominance of grasslands in the region and the different spatial configurations among the canton makes it a highly suitable case study to analyze the provision of multiple grassland ES in a multifunctional agricultural landscape.

3.2. Methodological approach

We proceeded in three methodological steps to characterize the provision of grassland ES from different grassland categories in a highly multifunctional agricultural landscape (Fig. 2). We focus on five

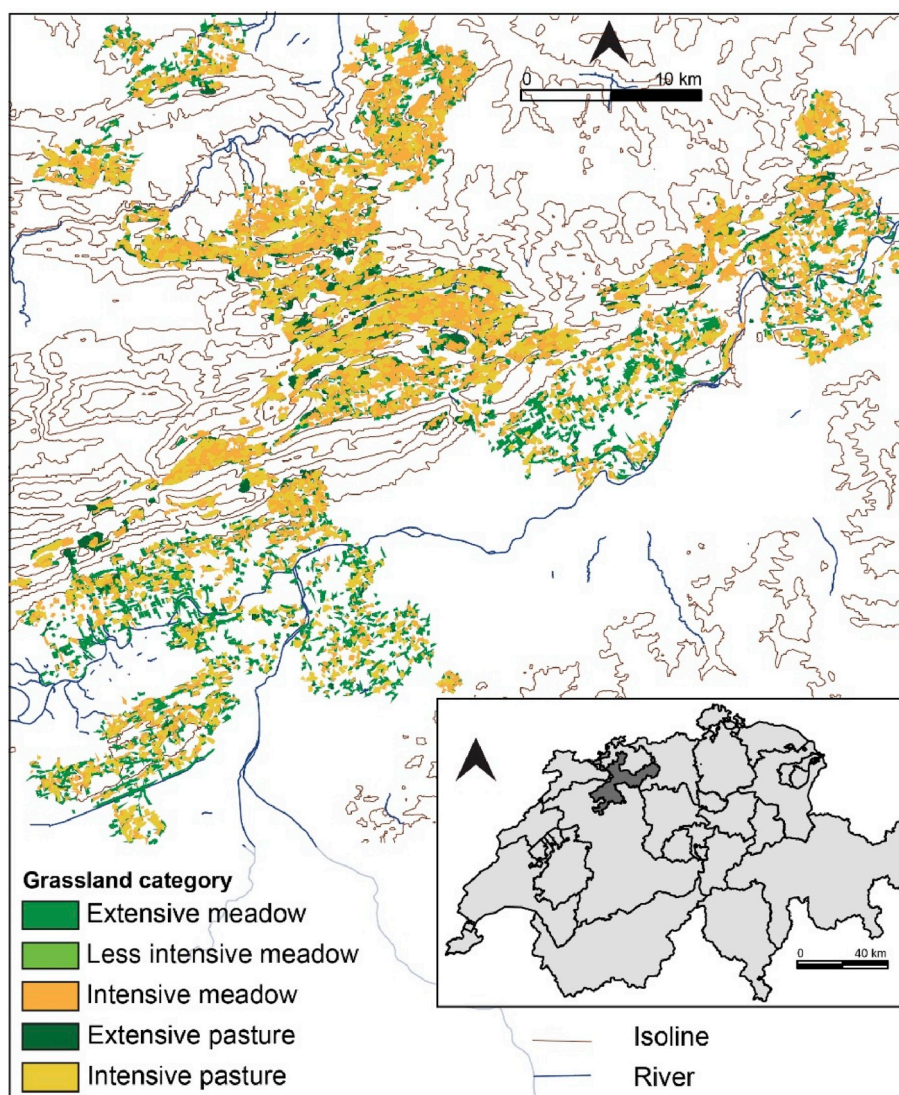


Fig. 1. Location of the Canton of Solothurn in Switzerland and of the grassland parcels within the cantonal territory.

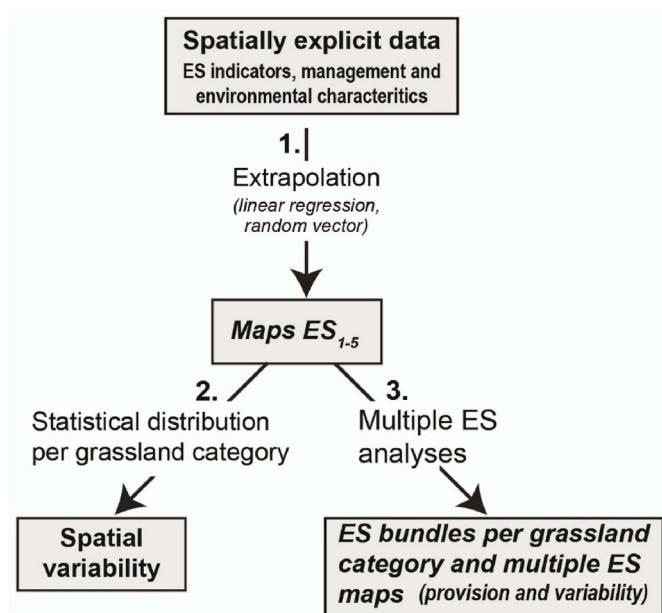


Fig. 2. Methodological approach to assess the provision of grassland ES.

indicators of ES provision (Table 2): yield (indicator of forage provision), Carbon (C) sequestration (composite indicator of climate regulation), bee species richness (indicator of pollination), vascular plant species richness (indicator of biodiversity conservation¹) and the number of photos taken in grasslands (indicator of outdoor recreation).

First, we used data on ES provision from various data sources to (1, Fig. 2) implement statistical approaches that extrapolate ES provision to our study area from a set of explanatory variables (Table 3). These statistical approaches differ from one ES indicator to another depending on the number of observations, the identification of potential explanatory variables in the scientific literature, and the nature of the collected ES data. For C sequestration, bee and plant richness, the relationship to the explanatory variables including management characteristics and spatial variables (Table 3) was modelled and parameters estimated with regression analyses. For the yields and the number of photos, we proceeded in one single step. For yields, we used the parameters of a regression that were estimated on the context of another study (Huguenin-Elie et al., 2017). We built a normal vector to map the number of photos.

Then, we assessed the spatial variability in the provision of the five individual ES by considering the difference between parcels in the different grassland categories (2, Fig. 2). We also considered the variation between the parcels and the mean provision under the same grassland categories. To do so we compared the modelled individual ES among grassland categories and we calculated the difference between the modelled individual ES provision of a specific parcel and the average ES provision of the grassland category of this parcel.

In the final step (3, Fig. 2), we used different statistical tools (Principal Components Analysis; PCA; and scoring) to analyze the provision of multiple grassland ES among and within the grassland categories. Analyses were done using the language and environment for statistical computing R (R Development Core Team, 2018) (R Core Team, 2018), with codes provided in the online Appendix. Below these three steps are explained in more detail.

¹ Biodiversity in grasslands can be considered a final ES *sensu* Mace et al., 2012. It provides important habitat services (de Groot et al. and can also be seen as a heritage value Rewitzer et al.

Table 2
ES indicators and data sources.

ES Category	ES	Indicator	Unit or range	Data collection	Sources	Type of data
Provisioning	Forage production	Yield	Ton of Dry Matter per hectare (t ha y ⁻¹)	Half-hourly measured fluxes, available for several years	Huguenin-Elie et al. (2017)	Modelled data based on field measurements
Regulating	Climate Regulation	C sequestration (composite variable)	Ton of C per hectare and year (t C ha y ⁻¹)	Aerial netting collection along a 2 × 100 m transect on three dates during good weather conditions in 2010	FLUXNET2015	Field measurements at 17 European flux towers
	Pollination	Bee richness	Number of species	One survey of 10 m × 10 m in 2010	Lüscher et al. (2016)	Field measurements from the canton of Obwalden (Switzerland)
Cultural	Biodiversity conservation	Plant richness	Number of species	17,979 photographs posted on Flickr between 2007 and 2017	Lüscher et al. (2016)	Field measurements from the canton of Obwalden (Switzerland)
	Outdoor recreation	Photos taken in grassland area	Number of photos between 2007 and 2017		Flickr (photo sharing social media website)	Data collected from the Internet

Table 3
Data used for the spatial modelling.

Data	Description	Source
Management	Regime Intensity Area	2 categories: Meadow or pasture 2 to 3 levels: intensive, less intensive (for the meadows only) and extensive Size of the parcel (ha)
Biophysical factors	Elevation Slope Soil	Average elevation of the parcel (m) Average slope of the parcel (%) Suitability for agricultural production. Five-classes typology based on slope, exposition and nature of bedrock
	Simpson diversity Patch Richness Distance to forest	Index of landscape diversity Index of landscape fragmentation Distance to closest patch of forest (m)
		Census data, Canton of Solothurn Aster Digital Elevation Model FOAG (2005) Corine Land Cover, 2012

3.2.1. Regression analysis

We implemented regressions for three individual ES indicators based on field data: C sequestration, bee and plant species richness. Such equations were necessary to identify the drivers of ES provision and to extrapolate ES values to the parcels of Solothurn based on the previously identified drivers. All these calculations were done at a parcel level.

3.2.1.1. C sequestration. To model the C sequestration (indicator of climate regulation), we considered direct measurements of the CO₂ net ecosystem exchange (NEE) in combination with C import to the grassland through fertilization and C export through harvest. We estimated the three components based on our data and on literature review.

In a first step, we estimated a linear function for the net CO₂ exchange (NEE) (based on flux data collected at flux towers in temperate grassland sites across Europe; the names of the sites are given in Appendix 1, see also section 3.4) for each parcel using three explanatory variables: management regime, management intensity and elevation as explanatory variables (Table 3; Eq. (1a)). A negative NEE meant a CO₂ uptake whereas a positive NEE reflected a loss. We selected the final model, through a variable selection procedure and estimated the parameters (α_0 , α_1 , α_2 and α_3). We employed Mallows' Cp as a goodness-of-fit measure (Mallow, 1973) to select best combination and identify the components of the management and of the landscape that affect the ES provision, and avoid noise and collinearity.² Such procedures are recommended to proceed to spatial extrapolations (Authier et al., 2017) and widely used in geostatistics (Hoeting et al., 2006).

$$NEE = \alpha_0 + \alpha_1 \cdot \text{Regime} + \alpha_2 \cdot \text{Intensity} + \alpha_3 \cdot \text{Elevation} \quad (1a)$$

with, NEE, net ecosystem exchange (t C/ha/year) and Elevation, its average elevation (in m).

In a second step, we estimated the carbon input (C_{input} through fertilization; Eq (1b)). The C input was calculated from two variables: the amount of recommended nitrogen fertilizers (N) spread on the parcel and the C/N ratio in the fertilizers. The amount of N fertilizers was based on the recommendation of fertilization by Huguenin-Elie et al. (2017). The C/N ratio of 6.1 was estimated from the data available for the observation sites used to model the NEE (flux towers across European grasslands) based on the amount of N and C contained in the organic fertilizers. The amount of recommended fertilizers applied depended on the yields (see below; eq. (3)). For more information, see Appendix 3.

$$C_{input} = N \cdot 6.1 \cdot \text{Yield} \quad (1b)$$

² We compared the results of the variable selection procedure using BIC and Mallows' Cp. When the results were different, we performed a cross-validation procedure. This procedure demonstrated Mallows' Cp to have a better goodness of fit for our data.

with, N, the recommended amount of nitrogen per unit yield (in t/t DM), Yield, the estimated yield (t DM/ha).

In a third step, we estimated the C exported out of the grassland systems ($C_{harvest}$ through harvesting; Eqs. (1c) and (1d) for meadows and pastures respectively). Due to the lack of data for the meadows, we used IPCC guidelines to estimate C export as 0.47 of the dry biomass (IPCC, 2006). For the pastures, we had information about the C exported for one of the Swiss observation sites.³ We built a linear model to link the yields and the C exported annually at this observation site and estimated the ∂_0 and ∂_1 parameters.

$$\text{For the meadows: } C_{export} = 0.47 \cdot \text{Yield} \quad (1c)$$

$$\text{For the pastures: } C_{export} = \partial_0 + \partial_1 \cdot \text{Yield} \quad (1d)$$

3.2.1.2. Vascular plant and bee species richness. We also implemented regression analyses to model bee and vascular plant species richness (indicators of pollination and biodiversity conservation, respectively; Eq. (2)). The models were built using observed data and a wider set of explanatory variables (Table 3): management regime and intensity, information about the landscape composition (i.e., size of the parcel, distance to the forest, patch richness, Simpson diversity of the landscape) and pedo-topographical features (i.e. elevation and slope).

$$\text{Species richness} = \gamma_0 + \gamma_X \cdot X \quad (2)$$

with Species richness, the richness in either bees or plants (i.e. number of species), X a set of explanatory variables about the management, its environmental characteristics and the characteristics of the surrounding landscape of the considered parcel (Table 3). We estimated parameters (γ_0 and γ_1) of the final model.

3.2.2. Extrapolation of ES values to the parcels of solothurn

3.2.2.1. Forage provision. To extrapolate the yield (indicator of forage provision) for each parcel in our case study region, we used the parameters of the linear models estimated by Huguenin-Elie et al. (2017). Yields for all grasslands were estimated depending of their regime, the intensity level considered as (β_0 and β_1) and elevation (Elevation; Table 3 and Equation (3)). Following Huguenin-Elie et al. (2017), below 500 m.a.s.l, yield estimations are equivalent to those calculated at 500 m.a.s.l. Above 500 m.a.s.l, elevation was used as a continuous quantitative variable. We also account for soil quality differences across space. The values presented above represent highest soil suitability (class one of a five-class typology). We used a correction factor (cf_p) to adjust yield estimates according to the information about soil suitability for agricultural production for each parcel (FOAG, 2005). For lower soil suitability classes (class 2 to class 5), the maximum yield is reduced by 5%–20%, respectively, representing assumptions made in other Swiss case studies (Mosimann, 2005).

³ Site of Chamau, for seven cattle grazing episode between 2001 and 2014.

$$Yield = (\beta_0 - \beta_1 \cdot Elevation) \cdot cf \quad (3)$$

with *Yield*, the estimated yield (t DM/year) and *Elevation* its average elevation (in m).

3.2.2.2. C sequestration. To map C sequestration, we extrapolated the NEE, the C input (through fertilization) and the C export (through harvest) to the study area by applying Equations (1a), (1b) and (1c) and 1d to the parcels of Solothurn. Finally, we calculated C sequestration for each parcel by accounting for NEE, C input and C export (Eq. (4)). To do so, we applied Equation (4), following Chang et al. (2015) to the parcels of grassland of Solothurn. A high C sequestration was the result of high C intakes (photosynthesis and/or C in fertilization) and low C losses (C content in harvests). For that reason, we considered the opposite of the NEE, as a negative NEE corresponded to a high CO₂ uptake of the grassland system.

$$C_{seq} = -NEE + C_{input} - C_{export} \quad (4)$$

with C_{seq} , the C sequestration (t C/ha/year), *NEE*, net ecosystem exchange (t C/ha/year), C_{input} the C imported in the system through fertilization (t C/ha/year) and C_{export} the C exported from the system through harvesting (t C/ha/year).

3.2.2.3. Plant and bee species richness. To extrapolate the bee and plant species richness, we applied the models that were estimated from the observed data to the parcels of Solothurn (Eq. (2)).

3.2.2.4. Number of photos. To extrapolate the number of photos (indicator of outdoor recreation), we used the photos taken in grassland in the Canton of Solothurn posted on Flickr. We assumed that all grasslands of a specific management strategy (i.e. the same grassland category) might provide similar outdoor recreation, as studies showed the influence of land uses on cultural ES, including outdoor recreation (Lindemann-Matthies et al., 2010). However, the elevation affects the provision of cultural services (Schirpke et al., 2017) and we considered here lowlands (< 800 m.m.a.l.s.) and highlands (≥ 800 m.m.a.l.s.) to model the outdoor recreation. We thus first subdivided the five grasslands categories into ten categories, using these two elevational levels. We extracted the number of photos taken in each of the ten grassland categories and years, between 2008 and 2017. Second, we built 'normal vectors' to extrapolate the number of photos. A 'normal vector' is defined by three parameters: its length, its average value and its standard deviation. We built ten normal vectors whose lengths corresponded to the number of parcels of the grasslands categories (management regimes, intensity and elevation levels). The average value of the ten normal vectors corresponded to the number of photos taken in 2017 per grassland category and elevation class. Vector standard deviations corresponded to the standard deviation of the time series per grasslands category and elevation class.

3.2.3. Estimating spatial variability in ES provision

The spatial variability of ES provision between parcels under the same grassland category in our study can originate from spatially explicit environmental factors (Parcel; Table 1) or on the characteristics of their surrounding landscape (landscape structure; Table 1). The spatial variability between parcels under the same grassland category originates from the variation in the primary data sources for each of the grassland categories. This type of variation was considered for all ES. Variations due to the landscape structure was considered for bee and plant species richness.⁴

To illustrate the range of the spatial variability on a landscape level, the modelled values can be presented in a density function showing the distribution curve of ES per grassland category in our case study region.

⁴ We did not find reliable data that accounted for the influence of landscape structure on the variation of yield, C sequestration and number of photos.

In addition, we identified the spatial variability by calculating the standard deviation of the individual ES under the same grassland category.

To identify the spatial variability within bundles, we calculated the variation Δ for each individual ES in each parcel. This variation Δ corresponded to the difference between the expected (modelled) ES $\hat{ES}$ value for a given parcel and the mean value of its corresponding grassland category $\overline{ES}_G$, (Eq. (5)).

$$\Delta = \hat{ES} - \overline{ES}_G \quad (5)$$

3.2.4. Identification of ES trade-offs, bundles and multiple ES maps

To characterize the provision of multiple ES, we performed five analyses at the parcel level based on the extrapolated data. First, we analyzed the trade-offs and synergies between the five individual ES indicators by performing a PCA. We used the extrapolated ES values for the 17,383 parcels (individuals) and all five individual ES indicators (variables), independent of management regime or management intensity. The PCA allowed us to determine the relationships among the five individual ES indicators. A constraint PCA allowed us to reveal the differences in terms of multiple ES provision between the five grassland categories, at the cantonal level and an s-class analysis was used to represent each category by its center of gravity and the links between each parcel and its specific grassland category.

Second, we performed five PCAs, one per grassland category, on the extrapolated ES values for the parcels (individuals) and all five individual ES indicators (variables). The number of statistical individuals depended on the number of parcels under the considered grassland category. The PCAs allowed us to bundles of ES that determine the relationships i.e., positive and negative associations among the five individual ES indicators between parcels under the same management. We projected the environmental factors, such as the elevation, as supplementary variables to understand their role in the multiple ES provision within the grassland categories.

Third, we analyzed the bundles induced by the management, by characterizing the multiple ES provision for each grassland category. For each grassland category and each ES indicator, we calculated the average modelled ES values ($\overline{ES}$) and performed a PCA on the new dataset. Next, for each grassland category and each ES indicator, we calculated the average of the standard deviation ($\overline{sd}$) of the modelled ES values and performed another PCA on the new dataset. From the two PCAs, we extracted the factorial coordinates of the projected variables (standardized $\overline{ES}$ and standardized $\overline{sd}$) and plotted them per intensity of management, for meadows and pastures separately, to get bundles of ES within the grassland categories. These bundles combine both average ES provision and its variability.

Fourth, we mapped the ES hotspots at the parcel level through the calculation of an overall score (Eq. (6)). We combined all individual-ES maps into an overall aggregated ES score based on the statistical distribution of each ES indicator (Lavorel et al., 2011; Le Clec'h et al., 2016; Maes et al., 2012). This overall score represents equal weighting across all individual ES indicators (*Score*). We transformed each of the five individual ES indicator into an ordinal score (Q_{ES}) from one to four, based on the quartiles (Petter et al., 2013) and summed up the five individual scores into an overall score.

$$Score = \sum Q_{ES} \quad (6)$$

Finally, we mapped the variation of the overall ES provision between parcels under the same grassland category at the parcel level. To do so, we standardized the individual variations Δ known for each parcel (see 3.2.2 for calculation of the spatial variability Δ in each parcel and each ES indicator). To standardize these variations Δ , we identified the minimal and maximal variation of each ES indicator for all parcels of the corresponding grassland category, $\min(\Delta_{ES})$ and $\max(\Delta_{ES})$. We used these minimal and maximal variations to get a

normalized value for each ES, comprised between -1 and 1 . Finally, we summed up the five individual variation Δ into an overall variation Δ_{total} (Eq. (7)).

$$\Delta_{total_p} = \sum_{ES} \left(2^* \left(\frac{\Delta_{ES_p} - \min(\Delta_{ES})}{\max(\Delta_{ES}) - \min(\Delta_{ES})} \right) - 1 \right) \quad (7)$$

3.3. Data

3.3.1. ES and their indicators

We studied forage provision through annual grassland yields. We followed the study by Huguenin-Elie et al. (2017) which combined parcel characteristics with empirical yield estimations. We studied climate regulation through the annual C sequestration based on net C exchange (NEE) as well as CO₂ imported and exported in the grassland system through fertilization and harvesting. We used in-situ measurements for a subset of grassland sites across Europe, the FLUXNET2015 dataset, for the calculation of climate regulation services (Pastorello et al., 2017). Bee richness is an important indicator to study pollination because almost all bees are pollinators and grasslands constitute an important habitat for them (Hudewenz et al., 2012; Meyer et al., 2017; Nogu   et al., 2016; Rogers et al., 2014). In addition, we studied biodiversity conservation through vascular plant richness in each of the grassland categories (L  scher et al., 2016). We used the Biobio dataset for the calculation of biodiversity conservation and pollination services (L  scher et al., 2016). Finally, to account for outdoor recreation, we used the number of geo-tagged photographs in different grassland categories per square kilometer and per year posted on Flickr. More information about the data sources and the development of the indicators can be found in Appendix 1.

3.3.2. Data for the spatial models

To administer the payments of different grassland categories, the Swiss Cantons collect spatially explicit census data on grasslands (and croplands) (GELAN, 2018). The census data comprise five categories of grassland management. These are based on two management regimes (pastures or meadows) and three intensity levels: extensive, less intensive and intensive meadows as well as extensive and intensive pastures (see Appendix 2 for details).

In addition, we used data on the biophysical characteristics of the parcels and of their surrounding landscape. Data about elevation were extracted from a Digital Elevation Model (DEM; 90m spatial resolution; Jarvis et al. (2008)) and slope (in %) was derived from the DEM. Information about the soil was published by the FOAG (2005). It constitutes a five-class typology based on the slope, the exposition and the nature of the bedrock. Information about the landscape composition and structure were derived from Corine Land Cover data, using landscape ecology metrics (<http://land.copernicus.eu/global/>). All information was extracted at the parcel level by calculating the average elevation and slope and selecting the soil class that covered most of the parcel.

4. Results

4.1. Identifying relevant drivers for modelling ES

Models differed in their ability to predict a given indicator but also in the number and nature of the variables selected (Table 4). Regime and intensity affect all ES indicators and their impact depends on the ES indicator. Elevation was an important driver of most of the ES indicators as well. It was negatively correlated with yield, NEE and plant species richness. For bee and plant richness, slope was also significant and negatively correlated. In addition, distance to forests was relevant to model bee species richness and was negatively correlated to the ES indicator, meaning that bee species richness increased with a decreasing distance to forest patches.

Table 4
Outputs of the statistical modelling: R², selected variables for each indicator. NA means that the model was applied but not built with primary data in the context of our study (yield) or that a regression method was not applied (outdoor recreation). Equations of the final models and coefficient estimates can be found in Appendix 4.

ES	ES indicator	Unit	Tested explanatory variables	Final model (significant explanatory variables)	R ²	N
Forage provision	Yield	t of DM/ha	NA (Huguenin-Elie et al., 2017)	Regime, intensity, elevation, soil	See notes ^a	83
	Climate regulation	t C ha ⁻² y ⁻¹	Regime, intensity, elevation	Regime, intensity, elevation	0.4	NA
Pollination	Plant richness	t C ha ⁻² y ⁻¹	NA	Yield and N fertilizer	NA	NA
	Bee richness	t C ha ⁻² y ⁻¹	NA	Yield	0.99 (pasture)	7 (pasture)
Biodiversity conservation	Plant richness	Number of species	Regime, intensity, elevation, slope, distance to forest, patch richness and Simpson index	Regime, intensity, distance to forest, slope	0.41	53
	Photographs	Number of photos	Regime, intensity, elevation, slope, distance to forest, patch richness and Simpson index	Regime, intensity, elevation, slope	0.75	53
Outdoor recreation			NA – application of a normal vector	Regime, intensity, elevation	NA	NA

^a For the three management intensity levels “intensive”, “mid-intensive” (which we did not consider here) and “less intensive”, 570 measurements were used from 120 sites (repetitions across years and/or botanical compositions within the sites). The equation were calculated using the mean yield measured on each site in order not to give more weight on the sites with more measurements. The overall R² was 0.827. For the management intensity level “extensive”, we used the yield estimation from (Dietl, 1986).

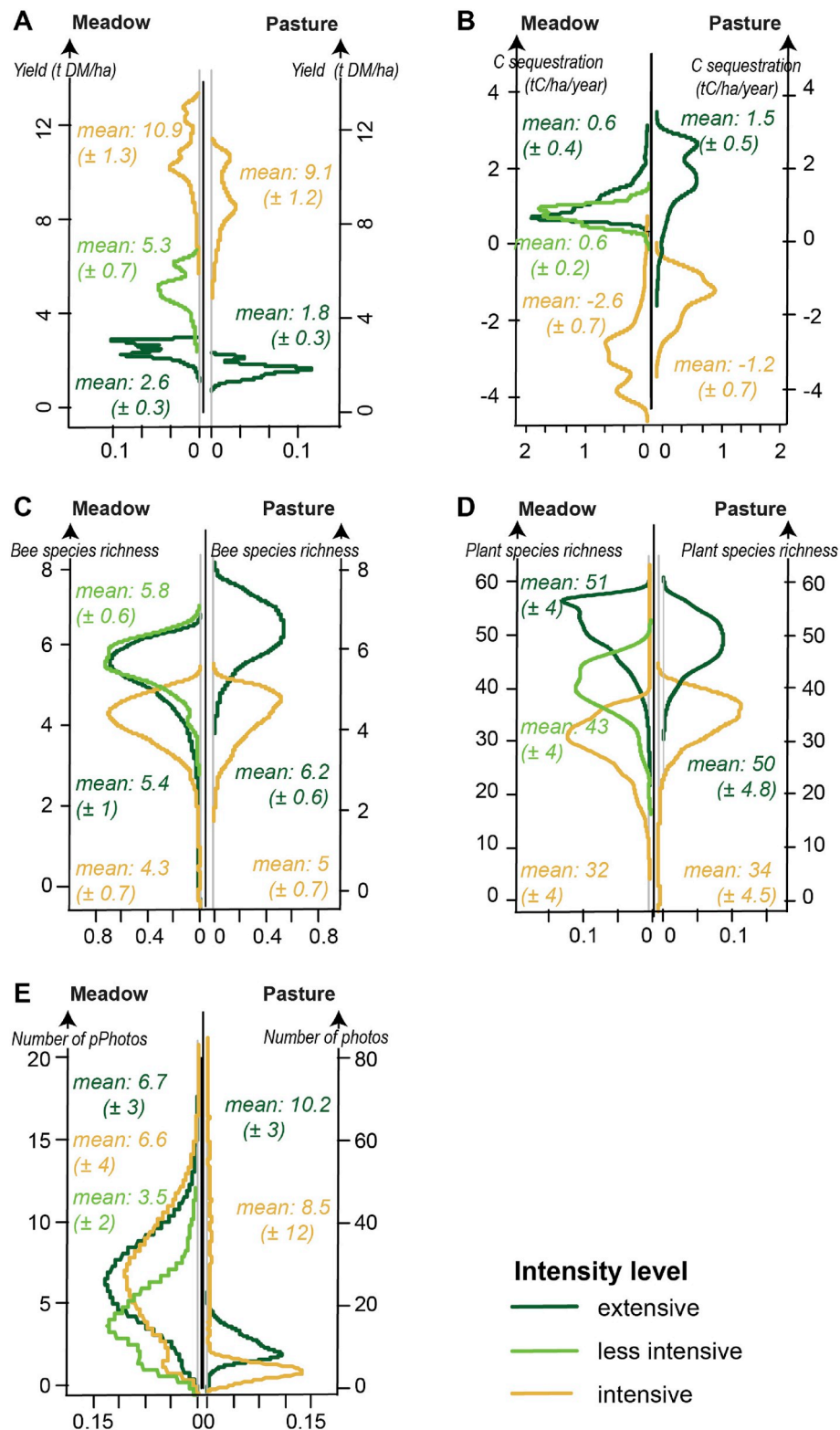


Fig. 3. Density functions demonstrate the probability of the modelled ES provision under a specific management (i.e. regime and intensity). Means and standard deviations (in brackets) are given. A. Grassland yields; B. C sequestration. Positive values reveal C sinks whereas negative values indicate C sources; C. Bee species richness; D. Plant species richness and E. Number of photos. Density functions that demonstrate the probability of the measured ES provision under a specific management are presented in [Appendix 4](#).

4.2. Modelling spatial ES provision

Based on the drivers of ES provision ([Table 4](#)), we modelled and extrapolated the five ES indicators to the parcels of the study area

([Fig. 3](#); Maps derived from the extrapolations are presented in [Appendix 4](#)). Kruskal-Wallis tests revealed that there were significant differences between the means of the ES indicators among grassland categories ([Table A5](#)). Modelled yields were significantly higher in meadows,

compared to pastures and increased with more intensive management (Fig. 3 A). In contrast, the average provision of other ES (C sequestration, plant species richness and in a lesser extend bee species richness) generally decreased with more intensive grassland land uses.

Despite spatial variability in C sequestration within grassland categories due to environmental factors, we observed a significant difference between intensive meadows versus all the other grasslands categories. Modelled C sequestration was on average positive in extensive pastures and in less-intensive and extensive meadows, implying that most of these grasslands categories were C sinks. In contrast, intensive grasslands were more likely to be C sources⁵ (Fig. 3 B). Modelled bee species richness was also higher in extensive compared to intensive grassland regimes (Fig. 3 C). There was, however, no difference between less intensive and extensive meadows. Modelled plant species richness also decreased with more intensive management and took on average slightly higher values in meadows than in pastures (Fig. 3 D). For number of photos, ES values were higher for extensive than for more intensive grassland, especially in pastures (Fig. 3 E). For meadows, we find no differences. However, the spatial variability between parcels under the same grassland category was much higher in meadows compared to pastures. Information revealed by the frequency distributions was critical to determine what level of ES provision to expect from a specific management. For instance, bee species richness is likely to range between six and nine for lowland pastures under extensive management, whereas it is likely to range between four and six under intensive management.

4.3. Revealing bundles, trade-offs and synergies within and across grassland categories

Fig. 4 reveals the relation between ES provision and its spatial variability between parcels under the same grassland category due to the underlying environmental factors. Across all intensity levels, higher provisions were related to higher spatial variability for yield as well as plant species richness in meadows. However, this pattern was not revealed for bee species richness nor for the number of photos, especially for the bee species richness for which more intensive meadows led to lower and more variable provision. Multiple ES provision varied strongly from one grassland category to another (Fig. 4).

Each grassland category exhibited a specific bundle of ES provision (length of the wedges). On average, intensive meadows were characterized by high yields and low plant species richness, bee species richness and C sequestration. Less intensive meadows presented medium ES provision for all indicators but for the number of photos, which was low. Extensive meadows were characterized by high plant species richness, a medium provision of C sequestration, bee species richness and number of photos but low yields. Similar trends could be observed for the ES provision in pastures. Intensive pastures were characterized by high yield, a high number of photos and low provision of the other ES indicators: plant species richness and bee species richness and C sequestration, whereas extensive pastures were characterized by low yield and high plant species richness, bee species richness and C sequestration.

The variability of the ES provision (width of the wedges) also depended on the ES and on the grassland category. In meadows, the overall variability of ES provision tends to decrease when going from intensive to extensive meadows. In intensive meadows, yield and bee species richness were very variable, whereas the number of photos and plant species richness had little variability. Yield, C sequestration, plant species richness and to a lesser extent, bee species richness, were relatively variable among less intensive meadows. The provision of all

ES was expected to be quite homogeneous among extensive meadows. Yield, number of photos and plant species richness, and in a lesser extent, C sequestration and bee species richness were very variable among intensive grasslands. Plant species richness and C sequestration were very variable in extensive pastures, whereas the provision of the other ES was not likely to change greatly among grasslands under this management.

PCA analyses performed for each grassland category revealed antagonistic and synergistic relationships between the five individual ES indicators, emerging from the underlying environmental factors (Fig. 6). They showed a very clear positive association between yield and plant species richness meaning that for a specific management regime and intensity, a parcel with high yields was likely to present relative higher plant species richness as well. The projection of the elevation on the correlations circle underlined the role of this environmental factor. Yield and plant species richness were likely to decrease in elevation. These two indicators were negatively correlated to the C sequestration, in all grasslands but less intensive meadows. Bee species richness was correlated with the axis 2 of the PCA and most of the time uncorrelated to the other indicators. The correlation of the number of photos with the other indicators varied from one grassland category to another. In extensive meadow, it was positively correlated with the bee species richness. In intensive pastures, it was positively correlated with C sequestration and negatively correlated with the yield and plant species richness. In the other categories, it was not well correlated with the two first axes of the PCAs.

The antagonistic and synergistic relationships between the five individual ES indicators, based on our PCA analysis, showed very clear trade-offs at the cantonal level (Fig. 5): yield vs. C sequestration, bee species richness as well as plant species richness (Fig. 5 A; axis 1). Number of photos could not be related as clearly to the other ES (axis 2). In terms of multiple ES provision, there was no clear distinction between the two regimes, at equal intensity level (Fig. 5 B). A clear shift exists between extensive versus intensive grasslands, irrespective of the regime. Even if there was a gradient from the extensive to the intensive grasslands, the proximity of the center of the scatter plots for extensive and less intensive meadows illustrated that the difference between these two intensity levels was tight, in the case of the selected ES indicators in our study region. The s-class analysis also highlights the considerable spatial variability in the interrelations between our ES indicators, as it was reflected in the dispersal of the dots (parcels).

4.4. Mapping multiple ES

Grasslands in the Canton of Solothurn exhibited hotspot areas of ES across all elevations (Fig. 7 A). Four trends could be observed in the provision of multiple ES in our study area. The overall ES provision (overall score based on the five individual ES scores) was, on average, higher in pastures than in meadows (Fig. 7 C). This overall score increased with less intensive land uses (Fig. 7 C and see Appendix 5). However, this result might be driven by the selection of our ES, as we modelled only one indicator of provisioning ES. The overall high ES provision was associated with high variation between parcels under the same grassland category (Fig. 7 B and C). Independently from the modelled level of overall ES provision, some parcels could be considered as hotspots because their modelled overall ES provision was higher than the average overall ES provision under current land-use in the study region (Fig. 7 B). Finally, the overall scores in extensive and less intensively used grasslands were mainly driven by regulating and cultural ES, whereas the score was mainly driven by provisioning ES in intensive meadows and pastures (Fig. 7 D). The overall ES score was almost equally driven by the yield, bee, plant species richness and number of photos, which each contributed on average around 20% of the score, though C sequestration contributed to ca. 17%. Decomposing the overall ES score is helpful to identify the potential of management options to provide multiple ES and to target and tailor specific political measures.

⁵ Negative value for intensively managed grasslands come from high exports which is a consequence of our assumption on C content of the biomass and on our data on yield.



Fig. 4. Bundles of the three intensity levels in terms of ES provision, A. for meadows and B. for pastures in the canton of Solothurn. The radius (length) of the wedges is proportional to the ES provision $\overline{ES}$ and the angle (width) is proportional to the standard deviation $\overline{sd}$. Wedges (length and width) were normalized through the PCA procedure.

5. Discussion and conclusion

In this study, we assessed the provision of five ES indicators and their variation between parcels under the same management practices in a multifunctional agricultural landscape. We also analyzed trade-offs and bundles induced by different management regimes and intensities. Our results highlight that the consideration of grassland categories, i.e., regime and intensity, is critical when assessing multiple ES and that information about land use only is not sufficient to model ES provision (Van der Biest et al., 2015).

We relied on a rich multi-source dataset. This dataset comprises census data as well as data on ES provision based on remote sensing and field measurement. Census data gives information about the real landscape and management practices at the parcel level. This means that our results are given for the spatial distribution of permanent grasslands in Solothurn in 2017. Due to change in management, ES provision is likely to change over time. The use of field measurements allows us to identify drivers of the ES provision, account for the characteristics of the region and to validate our maps, through statistical approaches. Our multi-source dataset allowed us to integrate diverse ES indicators, related to different ecosystems and ecological functions and components.

5.1. Associations and trade-offs in grassland ES

Understanding of relationships among ES is key to support a sustainable management (Zhao et al., 2018), especially in multifunctional agricultural landscapes (Frei et al., 2018; Manning et al., 2018; Raudsepp-Hearne et al., 2010). In this contribution, we showed that associations in grassland ES strongly vary among grassland categories leading to trade-offs in ES provision on a landscape scale. This finding is in line with other studies addressing the impact of grassland management on multiple ES provision (e.g. Briner et al., 2013; Divinsky et al., 2017; Kim et al., 2016; Schirpke et al., 2017; Wu et al., 2017). Our results are also coherent in terms of order of magnitude and trends across the management and environmental factors with other studies in Swiss agroecosystems addressing plant and bee species richness, C sequestration as well as for outdoor recreation (Dietschi et al., 2007; Junge et al., 2015; Kampmann et al., 2008; Le Feon et al., 2010). In addition to these studies, our results exemplify the spatial variability within different management strategies. We found that under the same management, flatter areas at low elevation tended to provide both relatively high yield and plant species compared to steeper areas on higher elevation (Wang et al., 2007).

Our results also reveal the spatial trade-offs between the

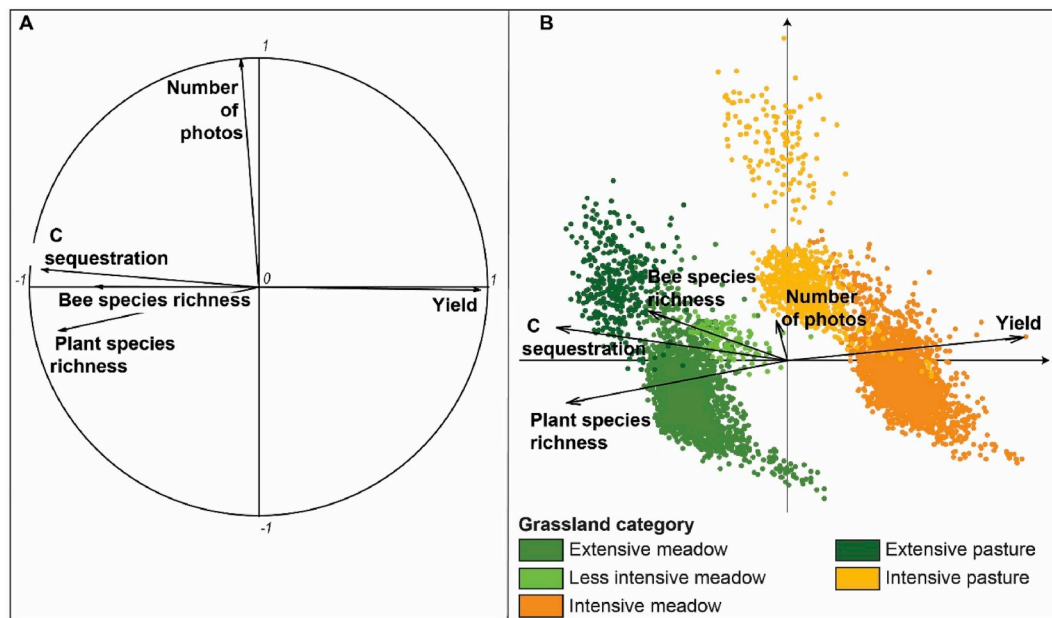


Fig. 5. Principal Components Analysis (PCA) and constraint PCA performed on the five ES indicators and the grasslands categories. A. Correlation of each ES indicator with the two first factorial axes is shown (arrows point in the direction of highest values of the ES). The first factorial map (axes1 and 2) of the PCA explains approximately 82% of total inertia (or of total variance of dataset). B. Factorial map associated to the correlation circle of the constraint PCA. Dots correspond to the parcels and are grouped by grassland category and their dispersal reflects the variability within the grassland categories; $R = 0.61$, p -value < 0.001). Pairwise correlations can be found in [Appendix 5, Table A6](#).

provisioning service (forage production) and most of the other grassland ES as shown in other studies addressing grassland intensities (e.g. [Allan et al., 2015](#); [Simons and Weisser, 2017](#)). These trade-offs between the provisioning and other services could be challenged by large-scale political strategies that would favor land sparing approaches ([Qi et al., 2018](#)). However, our results imply that there are exemptions depending on grassland categories and underlying environmental factors. In our case study, for example, yield and recreational services have positive associations in intensive but not in extensive meadows. Pollination services are highest in less intensive grasslands. This supports the finding that local associations might not scale up to the landscape level ([Qiu et al., 2018](#)) and that trade-offs can vary when considering multiple management options ([Beckmann et al., 2019](#); [Van Vooren et al., 2018](#)). Therefore, while nearly all ES studies to date do not focus on the parcel level, our results strengthen the importance of considering different management regimes and intensities at the plot level as the key level for managing trade-offs and synergies in grassland ES.

5.2. Spatial variability in ES provision

We also found that a higher overall provision increases the spatial variability of grassland ES. This does not only apply for provisioning services such as yield (e.g. [Finger and Buchmann, 2015](#)) but also for regulating and cultural ES. For plant species richness, climate regulation and bee species richness, a higher provision is also associated with higher variability in our case study region. We could not find existing literature that focused on the increase of variability with increasing levels of ES provision in grassland using empirical data.

Our results also highlight that while management practices are important drivers of ES provision and variability, they are not the only ones. As other studies previously showed, ES provision and their interrelations also vary according to biophysical and landscape components ([Simons and Weisser, 2017](#)). We found high spatial variability in regulating ES independent of grassland categories. In our study, some ES indicators, such as plant and bee species richness, are more influenced by landscape structure than by management practices. This is also true for the assessment of multiple ES, as existing research suggests

that there is considerable influence of spatial levels and dynamics over time on the assessment of trade-offs and synergies among ES ([Qiu et al., 2018](#); [Rau et al., 2018](#); [Sun and Li, 2017](#)). Our results exemplify that spatial characteristics in combination with different management regimes amplifies the variabilities in the provision of multiple ES. This implies that the method to extrapolate different data sources into a multifunctional landscape is challenging. Future research should also test different types of how to model and map variabilities in ES provision ([Andrew et al., 2015](#); [Lavorel et al., 2017](#); [Yang et al., 2018](#)).

5.3. Limitations

Our results underline that the consideration of grassland categories on plot level is critical when assessing trade-offs in ES. Data availability is a crucial limitation in this respect. For some grasslands categories, we could only find a small number of observations, e.g. regarding the bee and plant species richness in pastures. Consequently, variabilities must be interpreted with care. In addition, we could not find sufficient information on the temporal variability in ES provision. This highlights important data gaps in ES trade-off assessments ([Wong et al., 2015](#)). Our study underlines the need for systematic information about the temporal variations in ES assessments to allow future research to capture ES relationships across space and time.

Because the environmental factors might affect differently the ES indicator, the variability of ES provision within and across grassland categories critically depends on the assessed ES indicator. Identifying ES indicators is challenging, especially because such choices can affect the trade-offs revealed between the indicators ([Maes et al., 2016](#)). We carefully chose indicators based on scientific literature, data availability and expert knowledge. Our chosen indicators have two major strengths. Firstly, they are diverse in terms of ES categories, underlying ecosystem functions and potential beneficiaries. Secondly, despite the effort to sample the data, they allow a certain replicability of the method to other agricultural landscapes. However, our assessment comprises one indicator of provisioning service, whereas it comprises two of regulating and two of cultural services. This asymmetry influences our results when identifying hotspot areas with an overall ES score. We

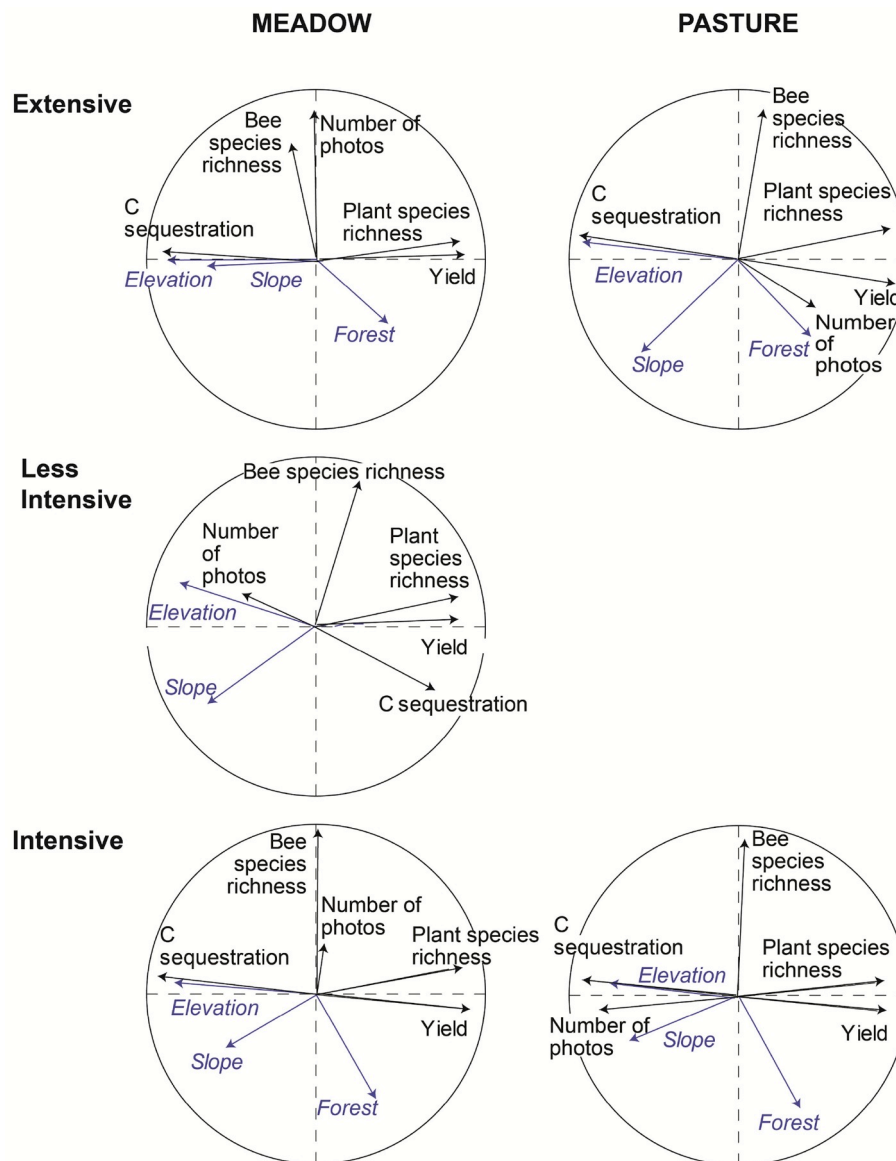


Fig. 6. PCA performed on the five ES indicators, for the five grasslands categories. Correlation of each ES indicator with the two first factorial axes is shown (arrows point in the direction of highest values of the ES). The first factorial map (axes1 and 2) of the PCA explains between 60% and 86% of total inertia (or of total variance of dataset, 74%, 60% and 74% for the extensive, less intensive and intensive meadows, respectively and 80% and 86% for extensive and intensive pastures, respectively), depending of the grassland category. In blue and italic font, the environmental factors, projected as supplementary variables.

chose to proceed with this asymmetrical design because of the lack of available data on forage quality. Considering the forage quality would have led to an assumption about the correlation between fertilizer and protein content. Because sward composition can be diverse under the same fertilization regime, such an assumption could generate high additional uncertainty.

5.4. Policy and management implications

The results from our analysis have two implications for policies addressing grassland management in European agriculture. Firstly, the spatial variability from underlying environmental factors would allow optimizing the spatial configuration of grassland management practices within a multifunctional landscape (Manning et al., 2018; Polasky et al., 2008; Simons and Weisser, 2017). The observation that forage production and biodiversity are positively associated within grassland categories, or management strategies, reinforces this potential since shifting grassland management in space could increase both ES

simultaneously.

The extent of the gains from re-allocation of grassland categories, however, is constraint by agricultural structures i.e., the prevailing small-scaled and family based farming systems in Europe. Our results thus clearly underpin the importance of cross-scale interactions in grassland ES management (Qiu et al., 2018). While there is increased interest in such landscape level approaches (Meyer et al., 2017), the successful implementation of such collective policies is challenging (Prager, 2015).

Secondly, the varying trade-offs resulting from different grassland regimes and intensities that we observe in our case study region implies that the support of low intensive grassland for biodiversity does not automatically improve the provision of other services (Allan et al., 2015; Frei et al., 2018). This had been shown also in other recent assessments of multiple ES (Birkhofer et al., 2018; Frei et al., 2018) and makes it difficult to design agri-environmental schemes in grasslands that address multiple environmental objectives (Galler et al., 2015; Schader et al., 2014). In this context, result oriented agri-environmental

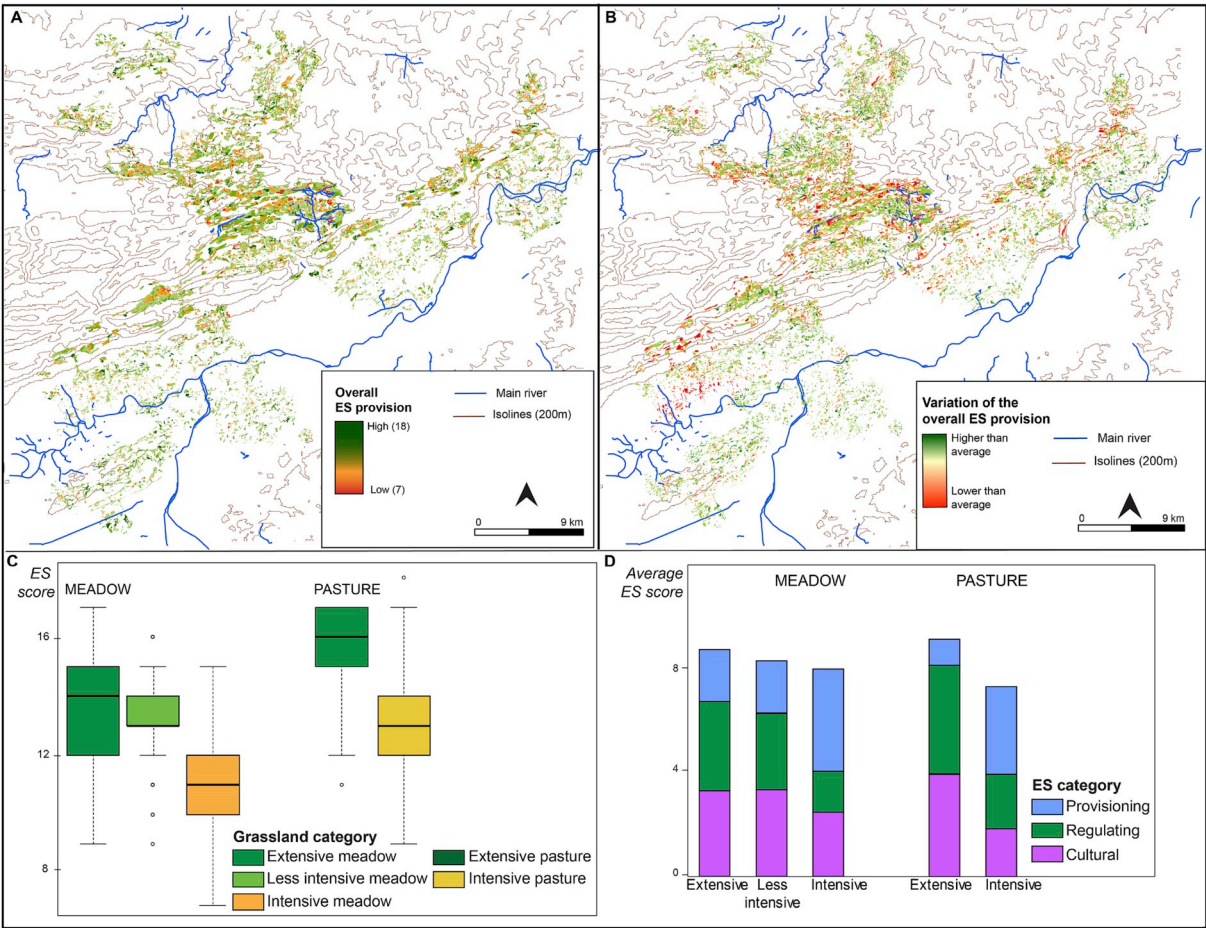


Fig. 7. Overall ES provision in Canton of Solothurn. A. Spatial distribution of ES hotspots in Canton of Solothurn. B. Difference Δ between the modelled overall ES provision of the parcel and the average value of the corresponding grassland category. Negative variation means the modelled overall ES provision of a parcel is lower than the average for the grassland category. C. Variation of the overall ES score within and between the grassland categories, and D. Composition of the overall ES score in terms of different ES categories. The provisioning, regulating and cultural categories are constituted of one, two and two ES indicators, respectively.

measures could provide more effective and cost-efficient incentives to provide grassland ES (Engel, 2016; Meyer et al., 2015). However, spatial targeting and payment differentiation are only partially applied in practice (Wunder et al., 2018) and many agri-environmental measures are still action-oriented measures, i.e. paying farmers for the delivery of input-reducing land management practices rather than the effective results (Burton and Schwarz, 2013). Thus, knowledge and information about spatial variability and the trade-offs from different grassland management practices as presented in our study will be of high importance for the design of effective and efficient agri-environmental policies in European agriculture.

Appendix

Appendix 1. ES variables

Yield

Type of data	Parameters of linear regressions
How data was acquired	Equations described in Huguenin-Elie et al. 2017

Such equations are crucial to estimate yields in the Swiss context if there are no field data available. The equations reported in Huguenin-Elie et al. (2017) were used to calculate the average yield for the elevation concerned in Switzerland. For each management regime and intensity level, an equation links elevation to the yields. Below 500 m elevation, the yield does not vary significantly with elevation; the estimated yield is therefore equivalent to that calculated for an elevation at 500 m. All equations are based on field work.

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Net CO₂ ecosystem exchange

Type of data	Quality-controlled ecosystem CO ₂ fluxes
How data was acquired	Field work
Data format	Raw
Experimental features	In-situ eddy covariance measurements at ecosystem scale globally
Data source location	Site locations are given with their official identifier in brackets. Switzerland: Chamau (CH-Cha), Früebühl (CH-Fru), Oensingen (CH-Oe1); Denmark: Enghave (DK-Eng), Rimi (DK-Lva); France: Laqueuille extensive (FR-Lq2), Laqueuille intensive (FR-Lq1), Lusignan (FR-Lus); Germany: Grillenburg (DE-Gri), Mehrstedt 2 (DE-Me2), Rollesbroich (DE-RuR); Hungary: Bugac (HU-Bug), Matra (HU-Mat); Ireland: Dripsey (IE-Dri); Italy: Monte Bondone (IT-Mbo); The Netherlands: Cabauw (NL-Ca1); UK: Easter Bush (UK-EBu)
Data accessibility	Public repositories, available at European Fluxes Database Cluster (http://gaia.agraria.unitus.it/home) and FLUXNET2015 dataset (http://fluxnet.fluxdata.org/data/fluxnet2015-dataset)

The Fluxnet database provides data on net ecosystem CO₂ fluxes and further atmospheric variables in grasslands (among other ecosystem types) around the world. Measurements of CO₂ fluxes can serve as inputs into models that predict the cycling of carbon, to detect the trends in climate, greenhouse gases and to understand how and why atmospheric state variables may vary across space.

Net ecosystem CO₂ exchange was measured using flux tower stations in temperate grasslands across the world. Some data were available online, other were requested from the person responsible for the respective station.

We used daily values derived from half-hourly data for the Net Ecosystem Exchange of CO₂ (USTAR Threshold; VUT) measured by eddy-covariance. An indicator of quality, ranging from 0 to 1, is associated with the data. We excluded the data whose values were below a quality control value of 0.66 and rejected the time series when it was not gapfilled. From the half-hourly data, we calculated the cumulative sum per year.

Bee and plant species richness

Type of data	Biodiversity survey
How data was acquired	Field work
Data format	Raw
Experimental features	Data were collected at 19 Swiss farms distributed along a gradient of management intensity and their elevation ranges from 605 m to 1137m.
Data source location	Canton of Obwalden, Switzerland
Data accessibility	Data published in Lüscher et al., 2016

Data about bee and plant species richness were collected within the EU FP7 Project Biobio research program (<http://www.biobio-indicator.org>) and were published in Lüscher et al. (2016). Species were collected in different habitats on the farms (including linear structures). In this study, we focused on grassland habitats solely. Farm management indicators for each farm were also collected (e.g., nitrogen input). Data about management were provided by farmers in face-to-face interviews following a standardized questionnaire.

Bee species richness. Data were collected at 19 farms located in the canton of Obwalden. The sampling was performed during the growing season in 2010, using standardized protocols (Dennis et al., 2012). Bees were sampled in a transect walk of 2 m x 100 m with aerial netting for 15 min, on three dates during good weather conditions, i.e. when conditions were dry and bright (cloud cover less than 50%) between 10.00h and 19.00h, with winds no stronger than Beaufort scale 4 (7 m s⁻¹) and temperature at or above 15°C.

Vascular plant species richness. Vascular plants were sampled using standardized protocols during one growing season, in 2010. The vegetation surveys were undertaken in plots placed at the center of the grassland to avoid edge effects. The plots measured 10 m × 10 m and were set up using survey poles with strings forming the diagonals of the square. Plots were orientated with the strings on the north-south and east-west axes. Vascular plant species were identified and recorded (except bryophytes and lichens). Their respective ground cover was estimated.

Recreation

Type of data	Georeferenced photos
How data was acquired	Flickr (a photo sharing social media website).
Data format	Raw
Experimental features	A total of 17,979 images were downloaded and used to analyse how many were present by grassland category and elevation, and further separated by year.
Data source location	Area of Solothurn, Switzerland.
Data accessibility	Flickr

Data from the photo sharing website Flickr is freely available online and has high potential for the assessment of cultural ES. Original images were uploaded to Flickr by users, often preserving the geographic location of where the photo was taken. The photo locations can thus be used as an indication of cultural ES, especially outdoor recreation. The data consists of the number of images that were uploaded within the grassland category between the years of 2008 and 2017 to Flickr for the study area of Solothurn, Switzerland.

The grasslands shapefile was projected from CH1903+LV95 to GCS_WGS 1984 using the CH1903+LV95_To_GCS_WGS_1984 geographic transformation in ArcMap v10.3. A python script was run to query the Flickr API at every 0.01 x 0.01 degree, capturing all points. The points were imported in ArcMap and used calculate points within each grassland and elevation types (with a 100m buffer) by year (between 2008-2017). Multiple buffer sizes between 25m and 150m were tested, before 100m was chosen, following Haider and Ali (2018), as they previously used this radius to gather Flickr data around the location of sites for assessment for cultural service assessment (specifically aesthetic). The number of resulting photos per km² were calculated using the original area covered by each grassland category, rather than the buffer area. A total of 17,979 photos were analyzed in this analysis. See Table A1.

Table A1

Photos by grassland category (per km²) by year. Calculated using a 100m buffer, with area calculation using original area size of each grassland category and two elevation levels. Lowlands are considered as grasslands below 800 masl and highlands as grasslands above 800 masl.

Intensity	Elevation	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Meadow											
Intensive	Lowland	3.64	5.06	1.95	4.22	3.74	6.18	9.54	15.61	7.35	6.64
	Highland	9.04	2.13	1.15	0.71	1.06	10.81	5.14	9.31	3.72	5.41
Less intensive	Lowland	1.75	2.28	2.46	1.40	1.23	2.10	7.72	7.37	2.81	3.33
	Highland	6.72	8.39	0.00	0.00	5.04	0.00	5.04	3.36	0.00	5.04
Extensive	Lowland	4.15	3.21	1.94	4.28	3.29	5.33	8.29	11.76	2.49	6.64
	Highland	0.88	5.97	12.46	2.81	4.74	3.16	5.44	10.00	3.86	7.20
Pasture											
Intensive	Lowland	4.44	3.92	2.38	1.89	2.01	6.35	6.57	6.38	2.83	3.85
	Highland	7.73	1.47	1.07	2.00	2.27	4.00	8.93	20.53	2.93	36.66
Extensive	Lowland	7.35	3.30	2.20	3.71	2.27	7.83	8.24	9.06	2.68	11.06
	Highland	14.50	6.40	12.24	3.77	37.49	13.37	18.08	9.04	2.83	7.91

Appendix 2. Description of the grasslands and the explanatory variables

Characteristics of the management (regime and intensity levels)

Parcels are varied in terms of management and environmental characteristics (Table A2). Census data were acquired from the Canton of Solothurn (<http://gelan.ch>) and are publically available. The data set provides georeferenced information across the Canton about the location of the parcels and some of their management characteristics, i.e., management regime and the level of intensity of their management. Moreover, the information has been validated by the farmers and is being used to distribute potential direct payments. We selected parcels of grasslands only.

The census data classify grasslands into five classes, based on their management: two regimes (pasture and meadow) and two to three intensity levels (intensive, less intensive and extensive meadows and intensive and extensive pastures). We used scientific literature to further characterize these five classes, according to their management (Blüthgen et al., 2012), e.g. in terms of amount of fertilizer, frequency of mowing or grazing. Here, we defined meadows as grasslands that are harvested predominantly by mowing over the last years or since sward establishment if it is younger than five years (Peeters et al., 2014). In agreement with the census set, meadows were divided into three levels of intensity: intensive, less intensive and extensive. Pastures were defined as grasslands that have been predominantly grazed over the last five years or since sward establishment if it is younger than five years. In concordance with the census database, we distinguished intensive from extensive pastures. We assumed that grasslands are well-balanced in species composition (i.e. they comprise 50 to 70% of grass; Huguenin-Elie et al., 2017). The use of these two parameters (management regime and management intensity) resulted in the establishment of five grassland categories (Table A2).

Elevation affects various variables related to environmental characteristics (e.g. species richness, productivity) and to management intensity (e.g. yield, number of cuts; Bergamini et al., 2001; Bühler and Schmid, 2001; Grandchamp et al., 2005; Güsewell et al., 2012; Jacot et al., 2000). We decided to distinguish two elevational classes. Despite an abundant literature, there is no consensus on thresholds to delimit elevational class. Based on previous studies, biogeographical knowledge and taking into account statistical constraints related to the need for a minimum number of observations per class, we decided to differentiate two elevational classes: lowlands (< 800m m.a.s.l.) and highlands (> 800m m.a.s.l.). We used this further distinction between grassland categories to model the recreational service.

Table A2

Distribution of the grasslands categories. In the lowland, many intensive meadows are not permanent grasslands but are included in a system of crop rotation and therefore we did not include these rotational meadows. When applicable, in brackets, the range of the values of the variables

	Meadows			Pasture	
	Extensive	Less intensive	Intensive	Extensive	Intensive
Number of parcels	6462	312	6900	1137	2572
Area (ha)	3350 (0.001-9.6)	156 (0.001-96)	6647 (0.0003-21.1)	1562 (0.01-25.9)	3764 (0.0003-23.1)
Elevation (m)	556 (324-1351)	585 (349-1269)	594 (305-1373)	705 (335-1370)	619 (325-1373)
Slope (%)	13 (0-87)	15 (0-56)	15 (0-87)	24 (0-75)	17 (0-77)
Distance to the forest (m)	226 (0-2300)	120 (0-985)	166 (0-2336)	93 (0-2234)	165 (0-2266)
Main class of soil suitability	1	4	4	5	5

Environmental characterization of the parcels and their surrounding landscape in the canton of Solothurn

The parcels of grasslands are georeferenced. Therefore, in a GIS, it is possible to calculate their area and to overlay their limits with other environmental variables to characterize the biophysical attributes of each parcel and its surrounding landscape (Table A3). To characterize the parcels, we used information related to topography (Aster DEM, available on <https://gdex.cr.usgs.gov/gdex/>). We calculated average elevation and slope, using the ArcGIS Spatial Analysis toolbox. We also used the soil classification (FOAG, 2005) to determine the dominant soil class. Landscape characteristics, distance to forests, Simpson's diversity and patch richness (metrics of landscape structure), were calculated from Corine Land Cover (available on <https://land.copernicus.eu/pan-european/corine-land-cover/clc-2012/view>).

Table A3

Characterization of the parcels of Solothurn, based on their environmental attributes and on the surrounding landscape (example of 20 parcels).

Regime	Intensity	Soil suitability	Slope (%)	Elevation (m)	Parcel Area (ha)	Distance to forest (m)	Simpson Diversity	Patch Richness
Pasture	Extensive	1	6	404	2.47	918.65	0	1
Meadow	Extensive	1	3	467	0.19	1215.46	0	1
Meadow	Intensive	5	20	972	5.35	443.54	0.26	2
Meadow	Extensive	3	6	479	1.00	1114.96	0	1
Meadow	Extensive	2	19	398	0.26	128.77	0.18	3
Meadow	Extensive	1	2	434	1.20	1632.10	0.27	2
Meadow	Intensive	2	8	630	0.55	806.07	0.23	2
Meadow	Extensive	3	2	436	0.26	1713.24	0.16	2
Meadow	Intensive	1	12	499	0.64	238.72	0.00	1
Pasture	Intensive	5	8	626	0.26	68.19	0	1
Meadow	Intensive	4	5	643	0.08	1985.28	0.43	2
Meadow	Extensive	1	5	460	0.09	1511.53	0.43	2
Pasture	Intensive	5	7	469	0.59	2051.49	0.00	1
Meadow	Intensive	5	10	590	0.09	1359.37	0.43	2
Meadow	Intensive	4	8	500	0.72	881.82	0.12	2
Meadow	Less Intensive	4	6	479	0.01	15.17	0.24	2
Meadow	Intensive	2	2	460	1.09	634.54	0.16	2
Meadow	Intensive	3	5	699	1.80	968.27	0	1
Meadow	Extensive	1	3	476	0.24	2458.71	0.25	2
Meadow	Extensive	3	7	668	0.20	698.75	0.	1

Appendix 3. C in fertilization

To calculate the C in fertilizers (C imported in the system), we first estimated the amounts of fertilizers applied on each parcel. We used the recommended amounts of nitrogen fertilizers presented in Huguenin-Elie et al. (2017) for each of the grassland categories (management regime * intensity) and per unit of yield. N fertilization recommendations ranged from 1.1 to 1.3 kg N/dt DM for intensive grasslands (meadows or pastures) and from 0.4 to 0.6 kg N/dt DM for less intensive meadows. Extensive grasslands are not fertilized at all in Switzerland. For each parcel, we thus multiplied the yield by the recommended amount of N fertilizer. Then, we applied the ratio C/N of 2.1 to reveal the amount of C contained in the organic fertilization. The C/N ratio was estimated from the data available from for two observation sites used to model the NEE (Chamau and Frübüel; Swiss flux towers).

Appendix 4. Additional results from the statistical and spatial modelling

Summary statistics of ES variables with management regime and intensity levels

The calculation of grassland ES for different grasslands categories showed the variation between regimes and intensity levels derived from empirical datasets (Table A4).

Table A4

Variations in measured ES indicators within the intensity levels of both meadows and pastures. We displayed the statistics for the time series of the indicator of outdoor recreation (number of pictures).

		Meadow			Pasture	
		Intensive	Less intensive	Extensive	Intensive	Extensive
Yield t/ha	Min	6.37	3.15	1.42	5.66	1.11
	Median	10.61	5.35	2.69	9.01	1.81
	Mean	10.84	5.32	2.64	9.15	1.82
	Max	13	6.4	3.05	11	2.4
	Standard deviation	1.34	0.67	0.33	1.28	0.29
C exchange t C / (ha * year)	Observations*	6411	303	5992	2433	1100
	Min	-1.7	-5.1	-2.8	-4.3	-4.4
	Median	-0.8	-0.9	-1.7	-2.4	-1.2
	Mean	-0.5	-1.3	-1.7	-2.5	-1.4
	Max	1.4	0.2	-0.2	-0.8	1
Bee species richness (number)	Standard deviation	1.1	1.7	0.7	0.9	1.6
	Observations	13	16	18	19	17
	Min	1	3	2	3	7
	Median	3	4	5	3.5	7.5
	Mean	3.6	4.8	5.1	4.0	7.5
Plant species richness (number)	Max	7	7	7	6	8
	Standard deviation	1.6	1.5	1.6	1.26	0.58
	Observations	17	11	12	6	7
	Min	22	29	36	20	44
	Median	26	38	48	32	46

(continued on next page)

Table A4 (continued)

		Meadow			Pasture	
		Intensive	Less intensive	Extensive	Intensive	Extensive
Number of photos	Mean	27.4	38.5	49.8	30.2	48.4
	Max	35	50	70	38	57
	Standard deviation	4.2	6.7	9	6.9	5.3
	Observations	17	11	12	6	7
	Min	0.71	0.63	0.91	1.17	2.23
	Median	8.31	4.2	6.82	4.39	7.02
	Mean	8.41	4.31	6.81	8.08	5.57
	Max	15.6	7.66	12.43	36.49	37.03
	Standard deviation	3.93	1.88	2.86	5.62	5.29
	Observations	20	20	20	20	20

* Because yield data were not measured on the field, the statistics summary for this ES was calculated for the grasslands of Solothurn.

Summary of the final models

To model the C sequestration, we first built a linear model based on CO₂ flux data collected at flux towers in temperate grassland sites across Europe to estimate the net exchange NEE_p (Eq. (1a)). We selected the final model, through a variable selection procedure and estimated the parameters

$$NEE = \alpha_0 + \alpha_1 \cdot \text{Regime} + \alpha_2 \cdot \text{Intensity} + \alpha_3 \cdot \text{Elevation} + \alpha_4 \cdot (\text{Intensity} \cdot \text{Elevation}) \quad (1a)$$

Multiple R²: 0.40; Adjusted R²: 0.35; p-value: < 0.001

Coefficients (α):	Estimate
(Intercept)	-0.7906
Regime-Pasture	-0.1441
Elevation	-0.0018
Intensity-Intensive	0.1414
Intensity-Less Intensive	-3.602
Elevation: Intensity-Intensive	0.0001
Elevation: Intensity-Less Intensive	0.0043

We also estimated the C exported from the grassland systems (C_{export} through harvesting; Eq. (1b) for pastures) for each parcel.

$$C_{\text{export}} = -0.01 + 0.41 \cdot \text{Yield} \quad (1b)$$

Multiple R²: 0.99; Adjusted R²: 0.99; p-value: < 0.001

Coefficients (β):	Estimate
(Intercept)	-0.01146
Yield	0.409002

Vascular plant and bee species richness. We implemented regression analyses to model bee and vascular plant species richness (Eq. (2)).

$$\text{Species richness} = \gamma_0 + \gamma_X \cdot X \quad (2)$$

with *Species richness*, the richness in either bees or plants, *X* a set of explanatory variables about the management, its environmental characteristics and the characteristics of the surrounding landscape of a specific parcel.

The final model for the bees presented the following variables:

$$\text{Bee Species Richness} = \gamma_0 + \gamma_1 \cdot \text{Regime} + \gamma_2 \cdot \text{Intensity} + \gamma_3 \cdot \text{Distance to the forest} + \gamma_4 \cdot \text{Slope}$$

Multiple R²: 0.41; Adjusted R²: 0.35; p-value: < 0.001

Coefficients (γ):	Estimate
(Intercept)	7.658406
Regime-pasture	1.139367
Intensity-Intensive	-1.243324
Intensity-Less Intensive	-0.079226
Distance to forest	-0.003152
Slope	-0.108794

The final model for the vascular plants presented the following variables:

$$\text{Plant Species Richness} = \gamma'_0 + \gamma'_1 \cdot \text{Regime} + \gamma'_2 \cdot \text{Intensity} + \gamma'_3 \cdot \text{Elevation} + \gamma'_4 \cdot \text{Slope}$$

Multiple R²: 0.75; Adjusted R²: 0.72; p-value: < 0.001

Coefficients (γ'):	Estimate
(Intercept)	62.592872
RegimePasture	3.186714
IntensityIntensive	-18.651679
IntensityLess Intensive	-7.149257
Elevation	-0.015758
Slope	-0.187069

Maps of the ES indicators

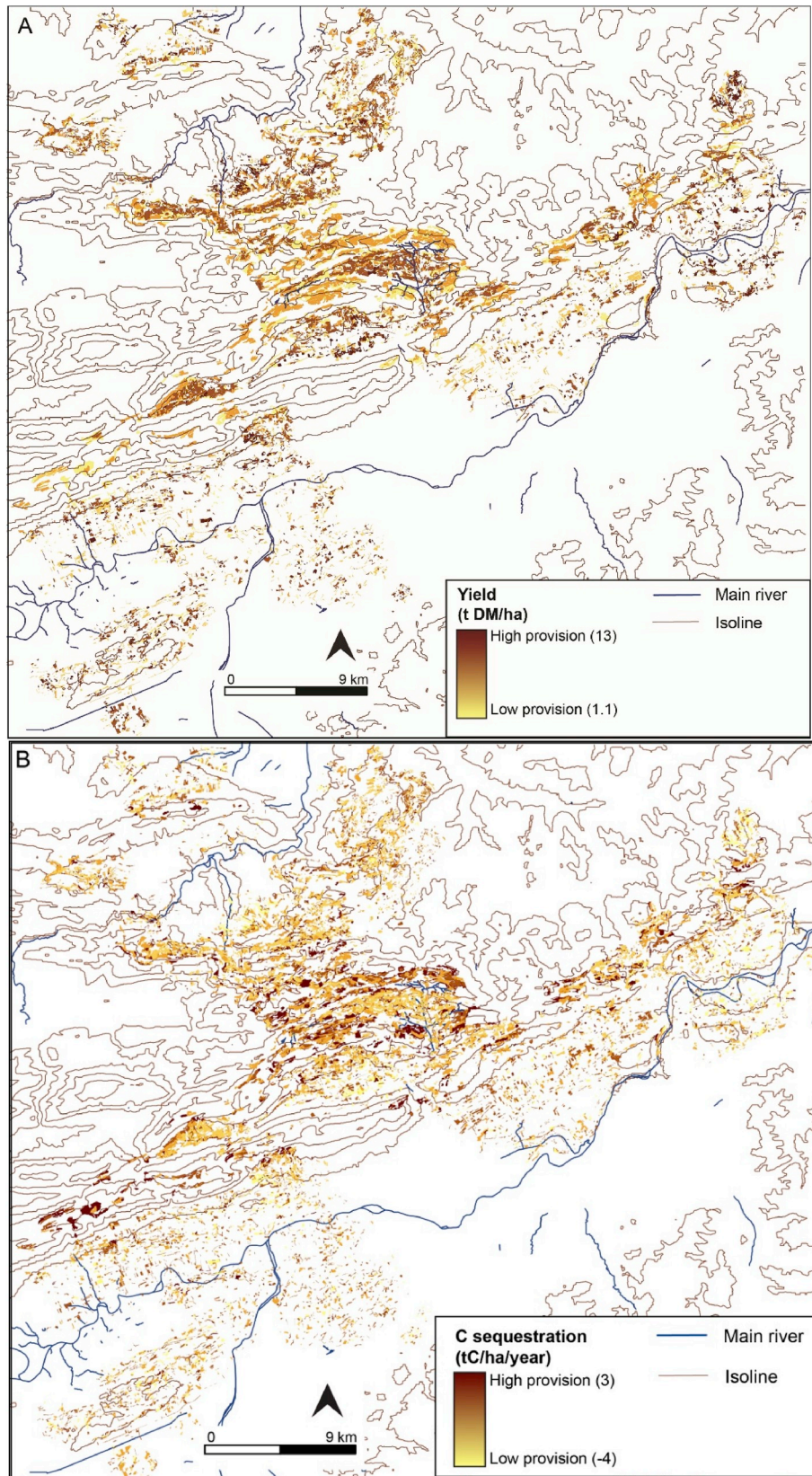
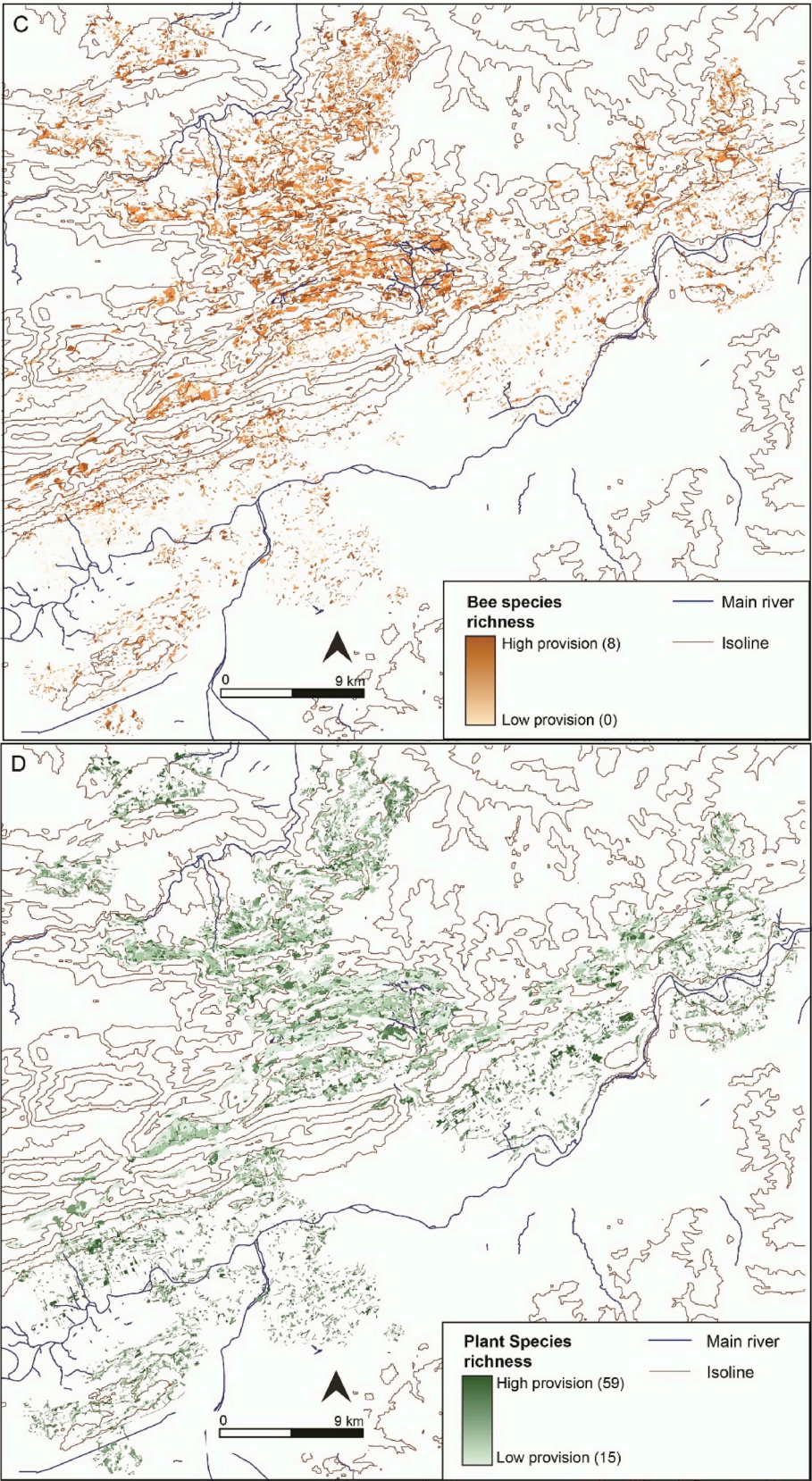
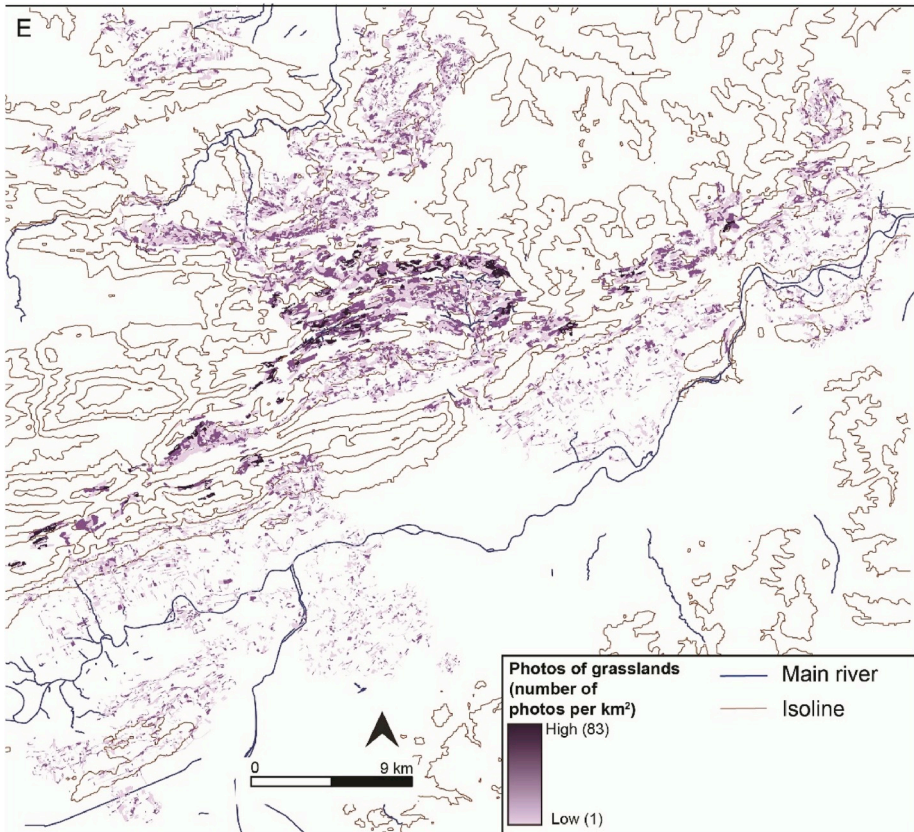


Fig. A1. Spatial distribution of the ES in Solothurn. A. Yield; B. C sequestration. Positive values reveal C sinks whereas negative values indicate C sources; C. Bee species richness; D. Plant species richness and E. Number of photos. In brackets, lowest and highest value of the ES provision.





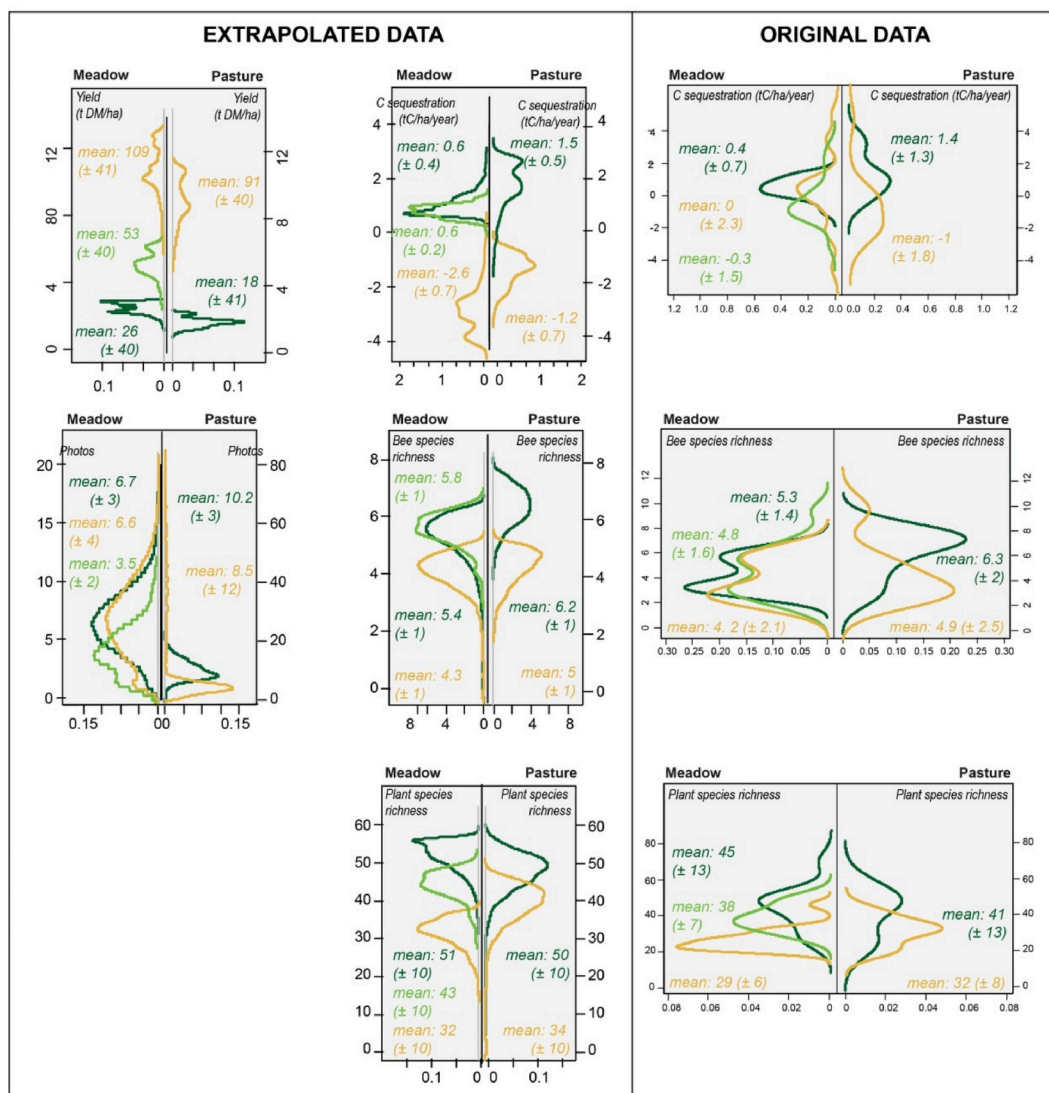


Fig. A2. Density functions demonstrate the probability of the expected ES provision under a specific management.

Table A5
Results of the Kruskal-Wallis tests.

	Kruskal-Wallis	
	Chi-squared	p-value
Yield	14251	< 0.001
C sequestration	14870	< 0.001
Bee species richness	8249.7	< 0.001
Plant species richness	13217	< 0.001
Number of photos	1926	< 0.001

Appendix 4 Additional results on multiple ES assessment

Table A6
Correlation table (Pearson's correlations) between the five modelled ES, for meadows and pastures.

MEADOW	Yield	C sequestration	Bee species richness	Plant species richness	Number of photos
Forage provision	1				
C sequestration	-0.99	1			
Pollination	-0.55	0.56	1		
Biodiversity	-0.85	0.77	0.49	1	

(continued on next page)

Table A6 (continued)

Recreation	-0.01	0.01	0	0.02	1
PASTURE	Yield	C sequestration	Bee species richness	Plant species richness	Number of photos
Forage provision	1				
C sequestration	-0.98	1			
Pollination	-0.58	0.55	1		
Biodiversity	-0.70	0.56	0.58	1	
Recreation	-0.24	0.33	0.01	-0.20	1

Table A7

ES overall score within the grassland categories, for all parcels of agricultural grasslands of the Canton of Solothurn.

Regime	Intensity level	ES score			
		Min	Max	Mean	Standard deviation
Meadow	Intensive	8	16	12	1.5
	Less Intensive	8	16	12	1.6
	Extensive	9	18	13	1.6
Pasture	Intensive	8	16	12	1.4
	Extensive	10	17	14	1.1

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2019.109372>.

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